The Corporate Infrastructure

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SERVERS

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INTRODUCTION TO SERVERS

What is a Server & their Role in a Network

In general, a server is defined as a specialized computer or a set of computers that runs the appropriate software and has the necessary resources to be able to provide services to other users, known as *Clients* over a network. A Server is named based on the function or service it provides. So, if it hosts webpages on the internet, it is called a Web Server, respectively, if it serves emails, it is called an Email Server etc.

Server function as the backbone of network environments be centralizing resources and ensuring efficient data flow between clients. They facilitate communication, data exchange, and resource sharing, making them indispensable for a network infrastructure. A Server can manage databases, host websites, deliver email services, store files, and run applications that users can access remotely.

Video: What is a server?

BASIC COMPONENTS OF SERVERS

HARDWARE

A server features a large Hard Disk capacity, powerful processing capabilities, a substantial amount of RAM, stable Internet connectivity, and fault-tolerant, high-quality components designed to continuous and reliable operation.

Server Chassis

It is exactly the same as the PC case for a commercial computer. It houses all server components: motherboard, PSU, hard drives, fans, NICs and more.

Furthermore, it plays a significant role according to system protection, cooling, and maintenance.

Type	Implementation
Tower server	Small Business
Rack-mounted server (or Rack server)	Data Centers
Blade Servers	Multiple servers in a share chassis (High-End)

Server Form Factors

Video: Different Server Form-Factors

Storage Devices

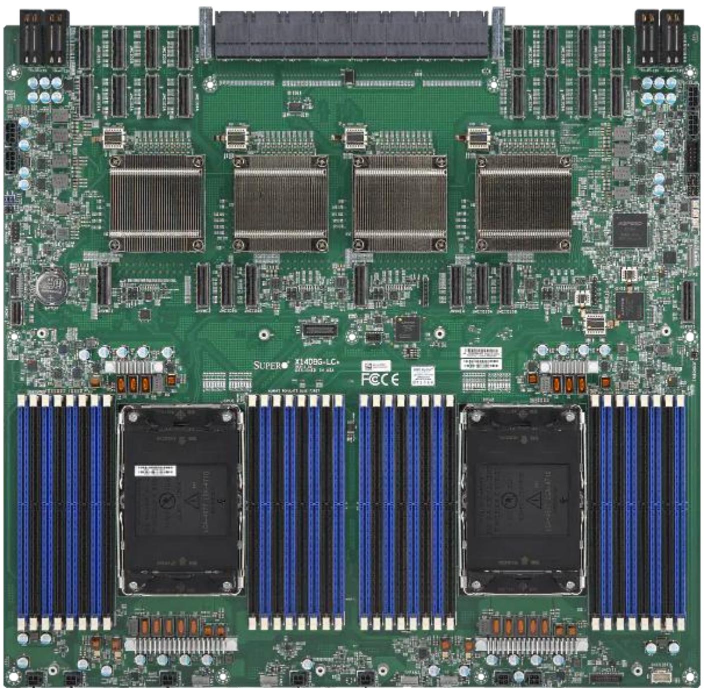
High capacity of Hard Disk Drives (HDD) or Solid-State Drive (SSD), for storing large amounts of data. SSDs are preferred for their faster data access speeds, which can significantly improve the performance of data-intensive applications.

Example: Servers are typically configured with **RAID** (***R**edundant **A**rray of **I**ndependent **D**isks*) technology to combine multiple physical hard drives into a single logical unit for security, redundancy, and performance improvement.

- Video: RAID Types , RAID 5 vs RAID 6 & What is Parity

Motherboard

The motherboard is the main component of every computer, or the Central Circuit Board and it is responsible to connect and coordinate all hardware components. The differences between a consumer-grade computer and a server-oriented MoBo are that the server motherboards are designed for high reliability, scalability, and continuous operation.



Video: How Motherboard works?

Video : Inside of a server explained

Random Access Memory (RAM)

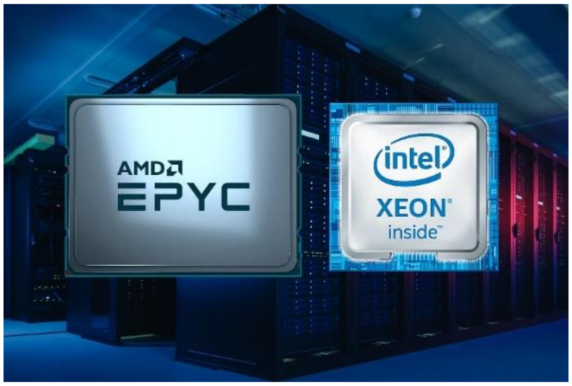
A server really needs to have massive RAM to manage multiple requests and processes simultaneously. More RAM allows servers to store and quickly access large amounts of data needed for efficient processing.

Example: Server often use *Error-Correcting Code* (ECC) RAM, which detects and corrects common types or data corruption.

Video: [ECC Memory]

Central Processing Unit (CPU)

High-performance processors with multiple cores and threads to handle multiple tasks efficiently. These processors are designed to provide the necessary power to manage large-scale operations and support high user loads without degradation in performance.



Example: Intel Xeon and AMD Epyc CPU are common in servers.

Video: How CPU works?

Network Interface Cards (NIC)

We have already explained what a NIC is, so on a server it is more useful to have multiple NICs for robust network connectivity. Redundant NICs can provide failover capabilities, ensuring network availability even if one connection fails.

Example: Dual-port or Quad-port NICs are common in servers for providing multiple network paths.

Power Supply Units (PSU)

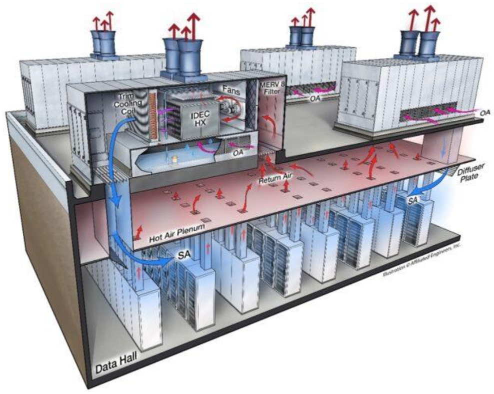
As with every computer, so in servers, the PSU provides energy to all server components. The key difference is that on servers, we always have to use redundant PSUs to ensure continuous power supply. Servers often have multiple PSUs, so that if one fails, the others can continue to power the server without interruption.

Example: Hot-Swappable PSUs allow replacement without shutting down the server.

Video: The PSU Connectors

Cooling Systems

In a Data Center, advanced cooling mechanisms such as liquid cooling or sophisticated air-cooling systems are used to prevent overheating. Efficient cooling is essential to maintain optimal operation and prevent hardware damage due to excessive heat.



Example: Data centers use advanced ***HVAC*** (**H**eating, **V**entilation and **A**ir **C**onditioning) systems to manage the heat output.

Video: Immersion Cooling

SERVER DEPLOYMENT: FROM LOCAL ROOMS TO GLOBAL CLOUDS

Right now that we have built the foundations of "What is a Server?" it is a good time to discuss the main models of hosting servers. We have to decide the implementation between a Server Room, Data Center, and Cloud Infrastructure, according to our company or organization's needs. The choice depends mainly on the organization's size, budget, and security.

Server Room

The Server Room is an in-house implementation on customer's premises. Is a dedicated room where all servers and networking devices are housed.

Usually Server Rooms are spanning in one or two rooms. Those room we call them server room, computer room or sometime machine room. In there we are trying to maintain Temperature and humidity in accepted levels to avoid any problem may occur. This can be managed through HVAC (Heat, Ventilation, and Air Conditioning) systems.

Server Room

How to Set up a Server Room

Data Center

A Data Center is a large-scale facility that hosts hundreds or thousands of servers for multiple organizations. Of course it offers advanced systems of cooling, power, security, and monitoring.

Data Center

Video: How Data Centers Work?

Cloud Infrastructure

Cloud Infrastructure is a hosting model where the resources (Servers, Applications, Storage) are delivered over the Internet. The key difference between hosting models is that in this implementation the users don't manage Physical Hardware but "consume" services:

The most well-known types of Cloud Computing Services are the following:

IaaS (Infrastructure as a Service) - (Microsoft Azure, Google Cloud, etc.)

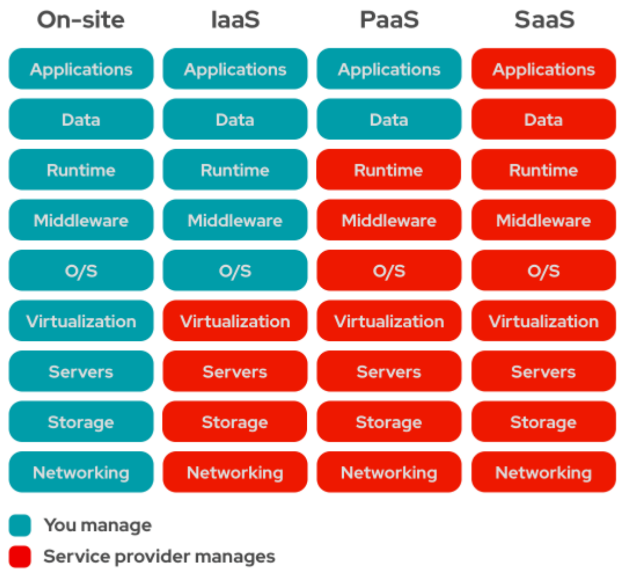
It is the closest thing to what we call on-premises infrastructure. In this case the cloud provider offers the client the part of Networking, Storage, Servers, and Virtualization. The customer is responsible for the rest.

PaaS (Platform as a Service) - (Google App Engine, Adobe Commerce, etc.)

The cloud provider, in this service, is responsible for almost the entire implementation, except for the client's applications and data.

SaaS (Software as a Service) - (Outlook, Gmail etc.)

In this case, the entire implementation is the responsibility of the cloud provider, which undertakes to offer its customer a complete solution in which the customer has no responsibility beyond Internet access.



On-Premises vs IaaS vs PaaS vs SaaS

Video: What is Cloud Computing?

Server Software

As we have already mentioned earlier, when we talk about servers, the first thought is about the Hardware: CPU, RAM, HDD/SSD, etc. But the real point is that without the right Software, a server is just a PC box full of expensive components. The Server Software is the entity that brings the server to life.

Servers are critical for the operation of modern information systems and networks. The software in servers plays a central role in resource management, application execution, security assurance and providing services to users. From operating systems to specialized applications, server software must be reliable, secure, and efficient.

Server Operating Systems (OS)

The Operating System is the foundation of any server. It provides the fundamental services for hardware management and application execution. There are various OSs designed for servers such as:

- Microsoft Windows Server
  - A popular Operating System from Microsoft, offering extensive support for applications and services like Active Directory and IIS.
- Linux
  - An open-source operating system with various distributions such as Ubuntu Server, CentOS, and Debian. It is known for its stability, security, and flexibility.
- Unix
  - An operating system which is known for its performance, and reliability, with distributions like FreeBSD and OpenBSD.
- ESXi/Proxmox
  - Specialized hypervisors for virtualization, without traditional OS layers.
- macOS Server (Discontinued 21 April 2022)
  - The server Operating System from Apple’s MacOS, offering seamless integration with other Apple devices and a user-friendly graphical management interface.

Server Applications

Software that enables the server to perform specific tasks such as hosting websites, managing databases, or running applications. Each server application is optimized for its role, whether it's serving web pages, managing email, or storing data.

Example: Apache server, Microsoft SQL server, Docker for containerized applications.

Management tools

Software for monitoring and managing server performance, ensuring security, and automating maintenance tasks. These tools help administrators monitor the health and performance of servers, automate backups, and ensure security compliance.

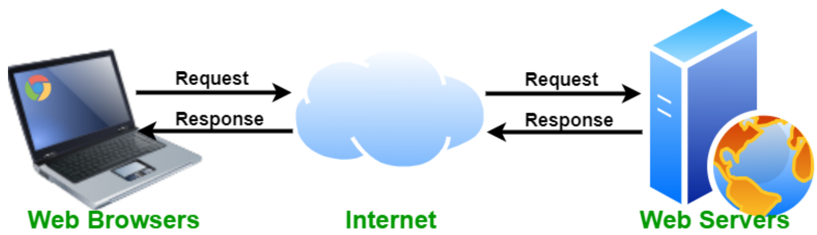
Example: Nagios for monitoring, Ansible for automation, and Splunk for log management.

SERVER TYPES & THEIR FUNCTIONS

WEB SERVERS

The function of a Web server is to handle the HTTP/HTTPS requests and serve web pages and content to users. So, web servers process incoming requests from users' browsers, collects the requested content, and send it back to the user.

Example: Apache HTTP Server, Microsoft Internet Information Services (IIS).



The Client requests a web content, and the Server sends it via Internet

Video: How Web Servers Work

Email Servers

It is the server that deals with the Delivery, Receipt, Routing and Storing of our electronic mail from the sender to the recipient. There are types of Mail Servers:

a) Outgoing Mail Servers

- **a.** *SMTP (Simple Mail Transfer Protocol)*:

This server is responsible to send an email from the sender's email ***client*** to the recipient's mail ***server***. It ensures that the email is correctly routed to its destination.

b) **Incoming Mail Servers**

- *POP3 (Post Office Protocol version 3):*

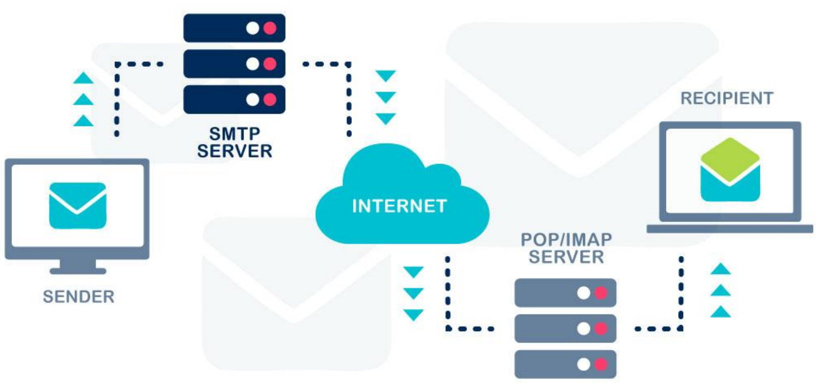
This server as soon as downloads an email from the mail server to the user's device typically deletes them from the server.

- *IMAP (Internet Message Access Protocol):*

This server allows users to access and manage their emails directly on the mail server. Emails remain on the server, and users can organize them into folders, mark them as read/unread, etc.

The email service is a very complicated procedure, let’s break it down in order to understand it:

- Step 1: Email composition
- Step 2: The sender’s email client communicates with sender's SMTP server and passes to it the sender's email, the recipients email, the text, and the attachments.
- Step 3: If the recipient is in the same Domain as the sender then the email is forwarded directly to the POP3 or IMAP server and from there to the recipient. In case the recipient is in a different Domain, then the SMTP server should contact other Domain Servers.
- Step 4: To find the recipient's Server the SMTP Server should contact the DNS Server, give it the Email Domain, and get the IP Address.
- Step 5: Once the SMTP Server knows the recipient's IP then it can communicate with the recipient's SMTP server.
- Step 6: The SMTP Server receives the email and checks it to see if it matches its Domain, then forwards it to the outgoing mail POP3 or IMAP Server for delivery to the recipient.



The email procedure

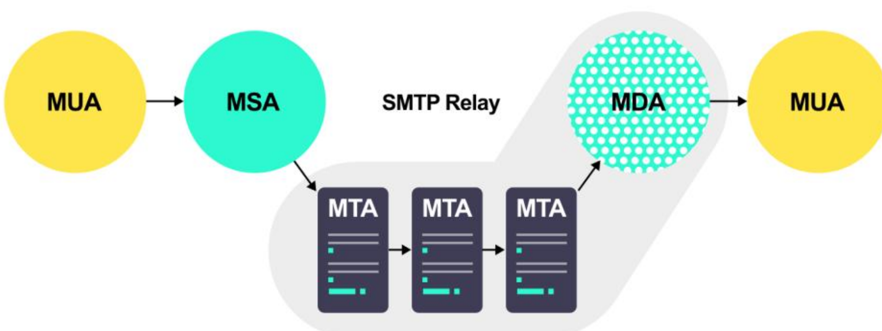
Video 1: [How eMail Works]

Video 2: [POP3 vs IMAP]

As we described earlier the email procedure is very complicated. Now we are going to discuss the components that work together to send, receive, and store emails.

- Mail Transfer Agent (**MTA**):
    - It is responsible for transferring emails from one server to another using the Simple Mail Transfer Protocol (SMTP).
- Mail Delivery Agent **(MDA)**:
    - It is responsible for delivering email messages from a Mail Transfer Agent (MTA) to the recipient's mailbox.
- Mail Submission Agent **(MSA)**:
    - Accepts outgoing mail from an email client and passes it to the MTA.
- Mail User Agent **(MUA)**:
    - Is the email client software used by the user to send and receive emails.
- Mail Storage:
    - It is responsible to store the email, until the recipient retrieves them.
- Domain Name System (**DNS**):
    - It translates the domain names into IP addresses to route emails correctly.
- Security Components:
    - Detect and block unwanted emails.

Example: Microsoft Exchange Server, Postfix.



The procedure

File Servers

They provide centralized storage and sharing of files across a network. File servers allow users to store, access, and share files from a central location, making collaboration easier and reducing data duplication and inconsistency.

Video: How FTP Works

Example: Microsoft Windows File Server, Samba (for sharing files between Linux systems and Windows systems.

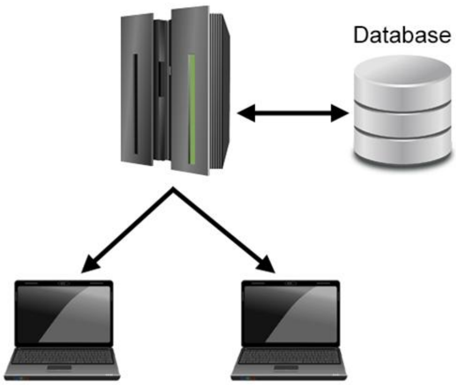
Clients can send, store, and restore files from the FTP Server

Database Servers

Host and manage databases, providing efficient data storage and retrieval. Database servers use relational or non-relational database management systems to organize, query, and maintain data integrity and performance.

Video: How Database Servers Work

Example: MySQL, PostgreSQL, Microsoft SQL Server.



The DB Server Stores and Maintains the DB that clients or applications use

DHCP (Dynamic Host Configuration Protocol) Server

It is the server that undertakes to automatically give an IP address, Subnet Mask, Default Gateway, and other network parameters to a device connected to the network. It reduces the administration workload, simplifying IP management and ensures that devices will communicate without problems within the network.

the process that DHCP follows to give the necessary information to the user is called **DORA**,

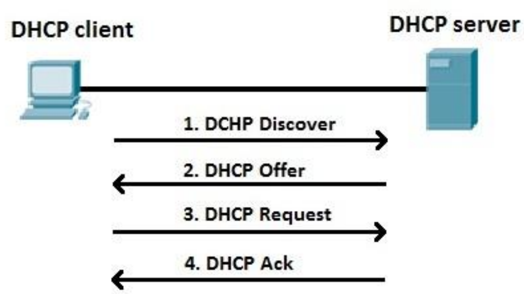
Discover, **O**ffer, **R**equest, and **A**cknowledge.

Discovery: *DHCPDISCOVER Message*, is a **broadcast** message send by a client who wants to join a network and tries to find an available DHCP server.

Offer: *DHCPOFFER Message*, is a **unicast** message send by the DHCP server as an answer to the DHCPDISCOVER message. In this message the server includes the IP Address, Subnet Mask, and all the other network parameters that the client needs in order to join the network.

Request: *DHCPREQUEST Message*, is the client's answer to the DHCPOFFER Message sent by the server and wants to indicate that the client has accepted the offered IP address, this is also a broadcast message to inform all DHCP servers on the network.

Acknowledge: *DHCPACK Message*, by this unicast message the server finalizes the whole process. This message confirms that the IP address has been assigned to the client and includes any additional configuration information.



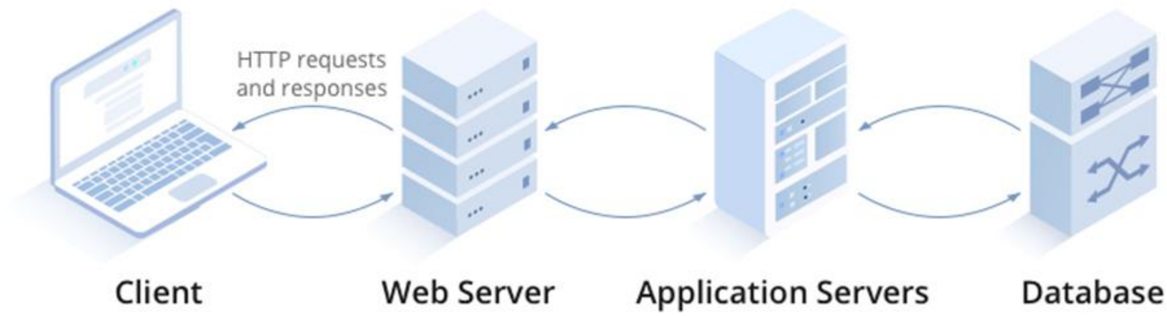
The DORA Procedure

Video: How DHCP Server Works

Application Servers

An application server is the type of server that hosts application services for users. Essentially, it acts as a middleman that manages the execution of applications, interactions with the database, and communication between users and applications.

Example: Microsoft Internet Information Services, Apache Tomcat, Oracle WebLogic, IBM WebSphere.

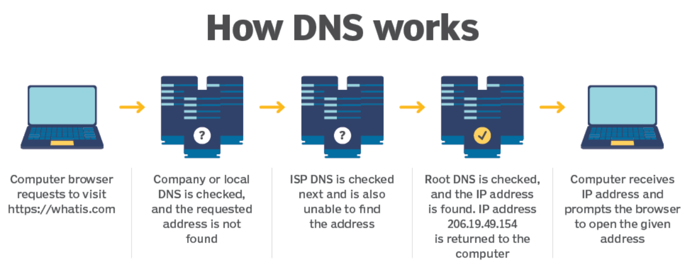


The Application Server acts a middleman

Video: What is an Application Server?

DNS (Domain Name System) Server

It is the server that has the IP addresses together with the Hostnames locally or accessible from the Internet. It is an essential element of the whole Internet infrastructure because it undertakes to translate the domain names we search for example in a browser, into IP addresses. For example it translates the "www.example.com" into the matching IP address. There is no DNS server that know every single IP. Instead, the Domain Name System (DNS) is a hierarchical and distributed system designed to efficiently manage the massive number of domain names and their corresponding IP addresses. *Root DNS Servers*



The DNS procedure for an unknown Webpage

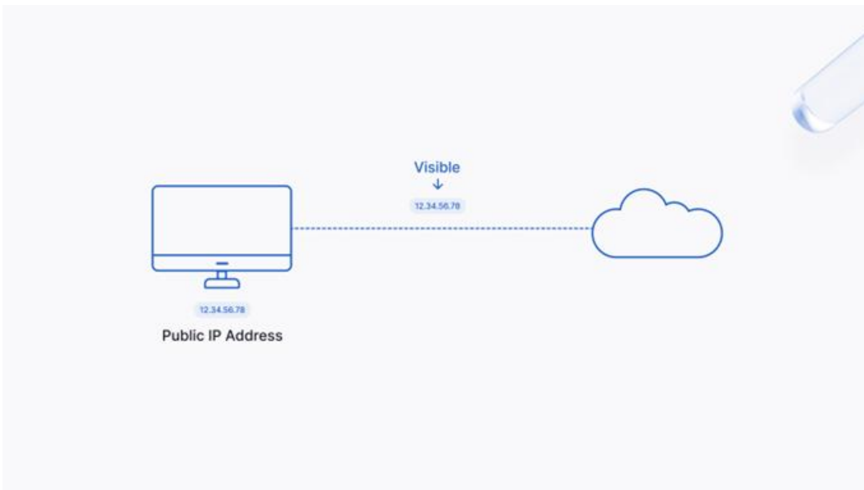
Video: How a DNS Server Works

Proxy Server

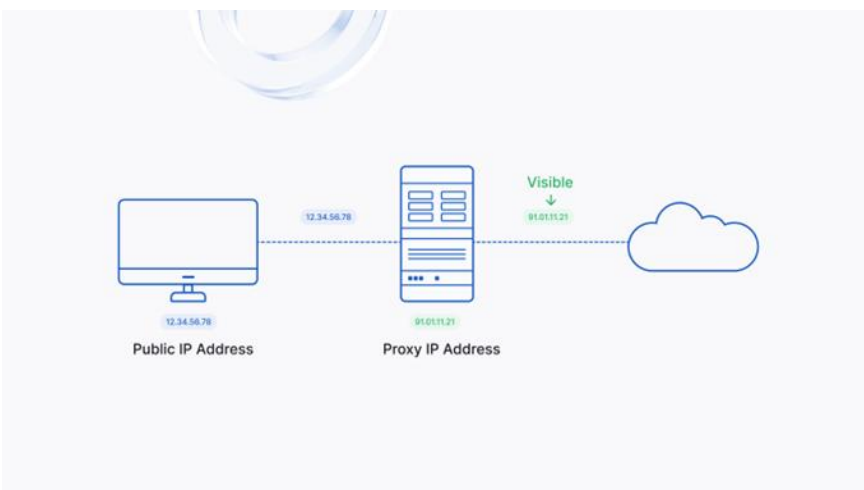
A proxy server acts as an intermediary between a local network and the Internet. It provides a communication interface between interacting networks by accepting requests from one network and forwarding them using its own IP address. These servers cache frequently accessed web pages, which helps them load faster when the user needs to open them again in the future. As a result, the network bandwidth is reduced considerably.

Moreover, proxy servers filter network communication and are always available, which is crucial for network load balancing. These servers also keep clients anonymous as their original IP address is replaced with a proxy.

Video: [What is a Proxy server?]



In this picture the Public IP of the user is visible to the Internet



The Proxy server acts as an intermediary between a local network and the Internet

EXTRA MATERIAL:

Video : Rackmount server anatomy

Video: Dell Server upgrade